Topological Quantum Memory

EE22B003, EP20B019

https://pubs.aip.org/aip/jmp/article/43/9/4452/230976/Topological-quantum-memory

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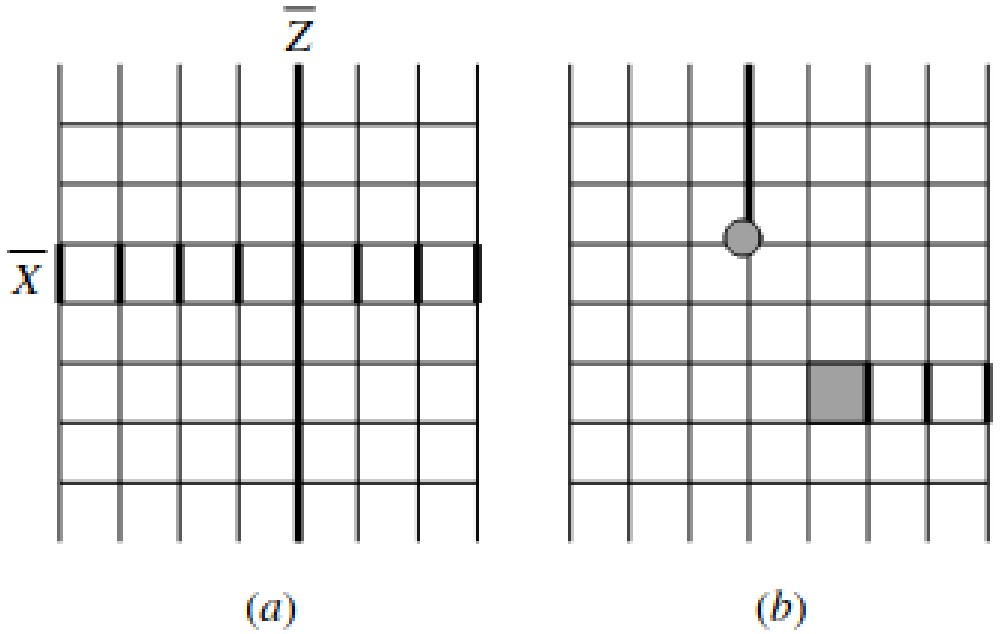
The Central Challenge: Protecting Quantum Information

- Quantum systems interact unavoidably with environment → decoherence
- Errors are **local and frequent**, but logical information must persist globally
- Standard QEC works, but requires active intervention and overhead

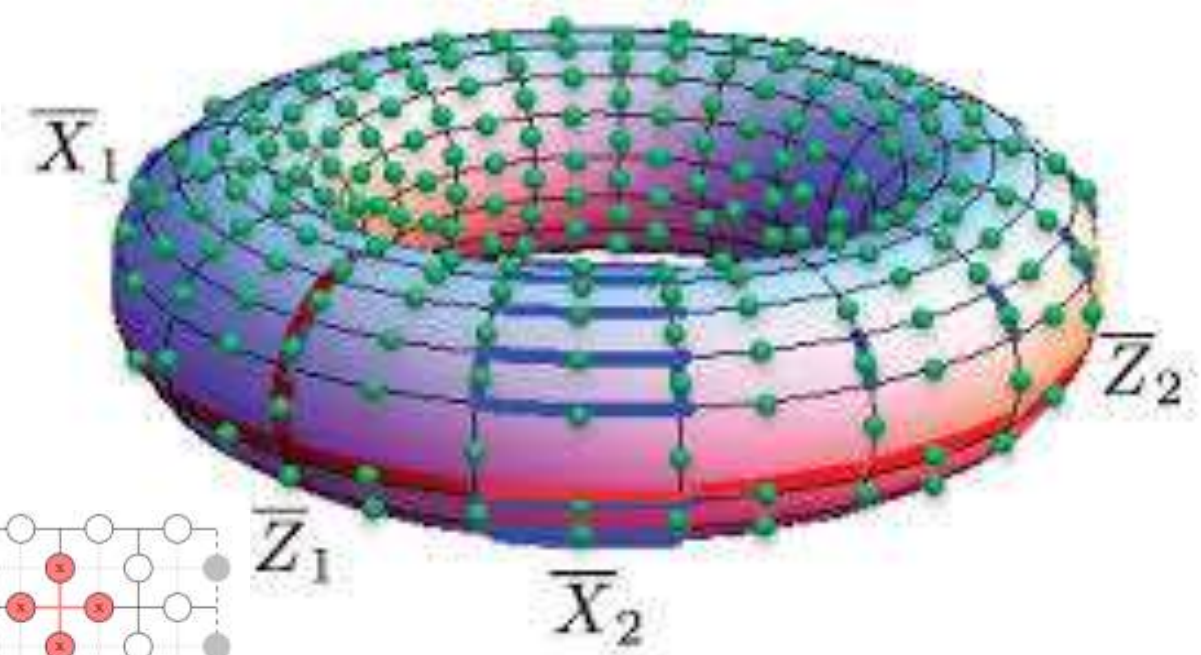
- Surface codes encodes information in **nonlocal, topological degrees of freedom**
- At the same time, make gates and syndrome generators local

Surface Codes

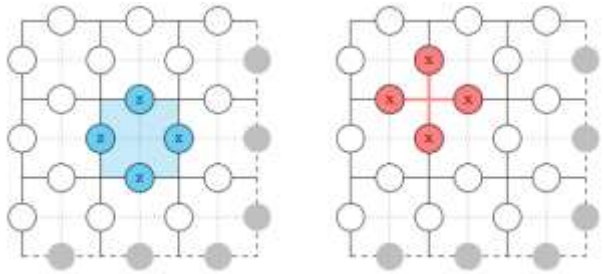
- Qubits arranged on edges of a lattice
- Stabilizer generators are local operators:
- Vertex checks (X stablizers)
- Plaquette checks (Z stabilizers)
- Errors always occurs in pairs (note
- that in 2D code, assume X error and
- Z error have probability p, Y has p^2 due to planar geometry.
- In this paper,
- Topological codes are realized in two ways, toric and planar

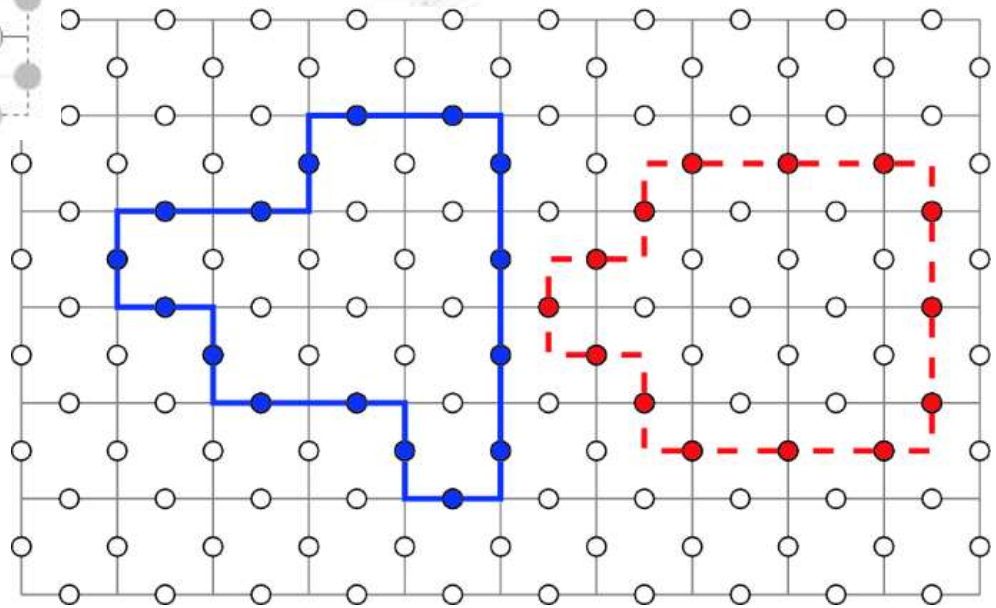


Toric code

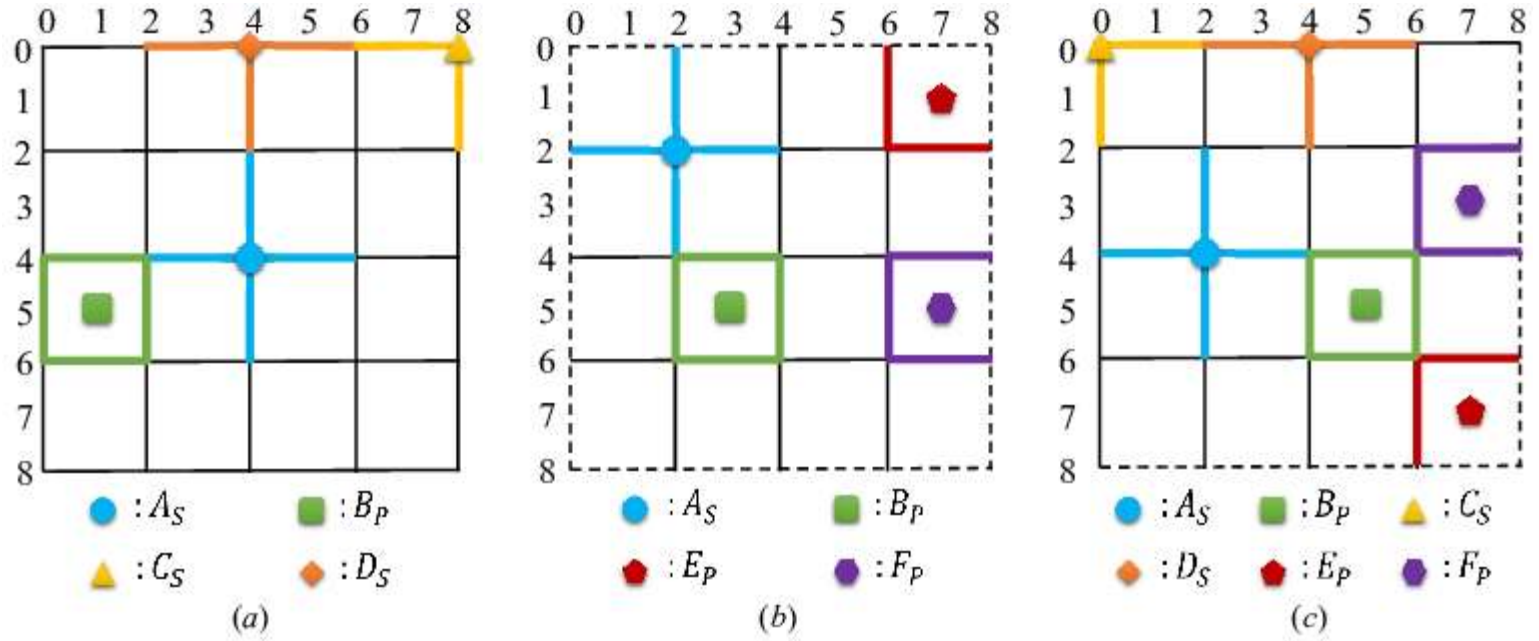


- Lattice embedded on a torus
- Consider an $L \times L$lattice on a torus
- Total no of qubits,$n = 2L^2$
- **Stabilizers:**
- One **X-type (star)** stabilizer per vertex
- One **Z-type (plaquette)** stabilizer per face
- Counts:
- Number of vertices = $L^2 \rightarrow L^2$,X stabilizers
- Number of plaquettes = $L^2 \rightarrow L^2$Z stabilizers ,Not all stabilizers are independent:
  - Product of all X stabilizers = identity
  - Product of all Z stabilizers = identity
- 👉 Independent stabilizers:$(2L^2 - 2)$
- **Encoded qubits:**
- $k = n - rank(\mathcal{S}) = 2L^2 - (2L^2 - 2) = 2$
- So the toric code encodes **2 logical qubits**





Planar Code (Physical Implementation)



- Toric Codes are hard to realize , physically, so we use planar codes with open boundaries in real systems
- Two boundary types : Smooth, Rough
- logical qubits encoded via strings connecting boundaries
- It stills retains the error correcting properties of toric code, although it has slightly reduced symmetry compared to toric geometry

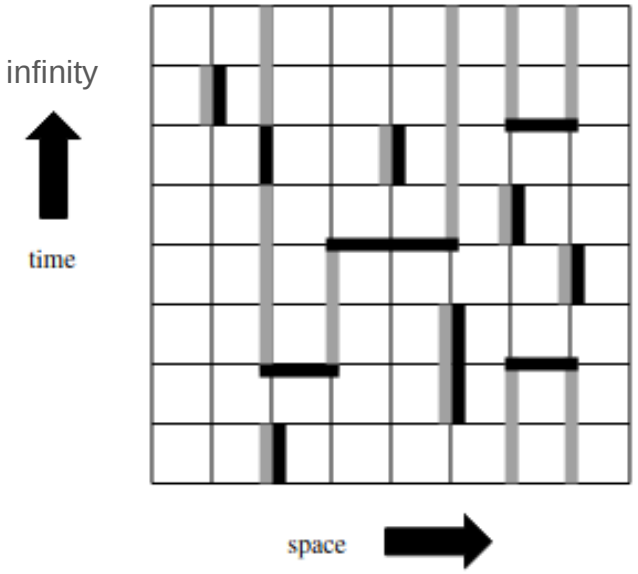
Intuition for Error Recovery



False syndrome bits



Flip bits above

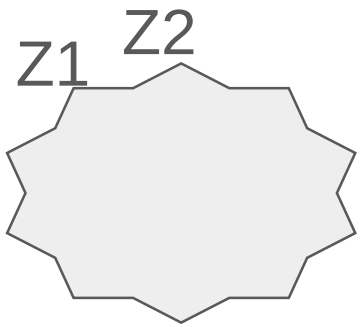




Measure no error, error is there



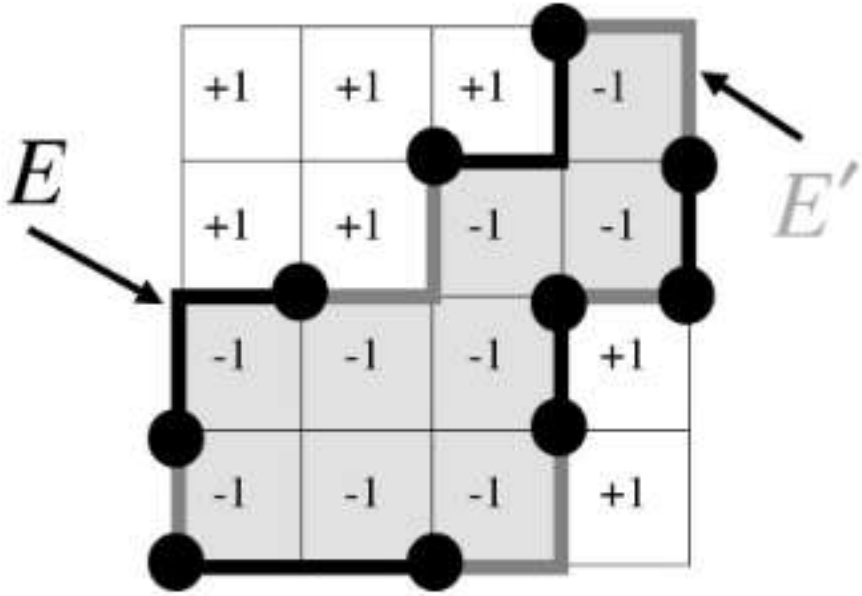
Correct syndrome measured



For visualization, the paper considers 1D repetition code

Some points -

- Most likely error chain is the one with minimum weight, but that may not be the safest strategy.
. What matters is the combined homology class of all the curves (Domain Walls) put together only.



Formulation of Fault tolerance theorom

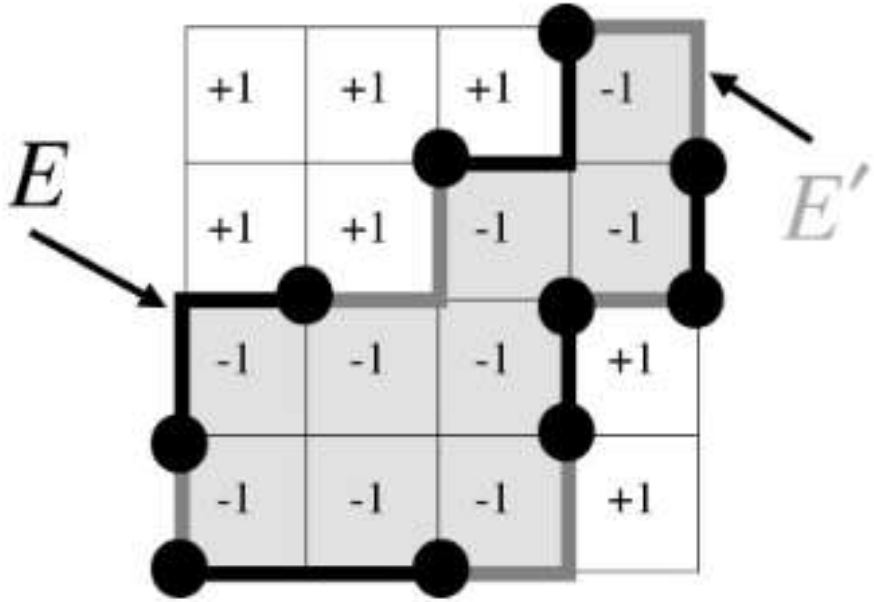
There exists a constant error rate pc (~11%) such that:

- If the physical error rate p<pc , then
  the probability of logical error after optimal decoding decreases **exponentially in system size (L)**.
- If p>pc, then the logical error probability is bounded away from zero (fails to be fault tolerant).
- Further, there exists pairs (c,q) where q is measurement error such that an optimal decoding taking potentially infinite amoun
  of time is almost sure to correct the error (the paper analyses the case q = pc , pc `1.14%)

Some things to note :

- The theorem does not exactly give a construction for the optimal decoding (that is harder than NP, but well approximated by
  MWPM)
- Exponential in L, not L^2 indicates percolation like behaviour (Pierls argument)

Connecting Fault Tolerance of Toric Code to Random Bond Ising model



Consider C to be the set of closed curves (all of

them together), and a fixed E (not S)

That contain S, the boundary of E.

Assign probability proportional to e^(-2*J*|E+C|) to C, J is a

parameter, |E+C| is calculated by adding along the curve, +1 when on E, -1 else.

e^-2J = {(p/(1-p) for a link not on E, (1-p)/p else, in optimal decoding as prob(E) proportional to p^|E|*(1-p)&(L^2-|E|)}. Now this is proportional to e^(-J*ul), where ul is - or + 1 depending on if it is on curve C or not (the factor 2 goes away as all L^2 like are used). (this can be rewritten as ul = Si*Sj where S are classical spins sitting on dual lattice next to each link (same algorithm for X and Z). This makes C appear like a domain wall in an ising system.

Finally, we make a partition function out of this, as Z[J,n] = Sum{over all Sij}(exp(J*Sum(nij*Si*Sj)), nij introduces a “random strength to the bond J, as 1 with probability p and -1 with probability 1-p).

Phase diagram

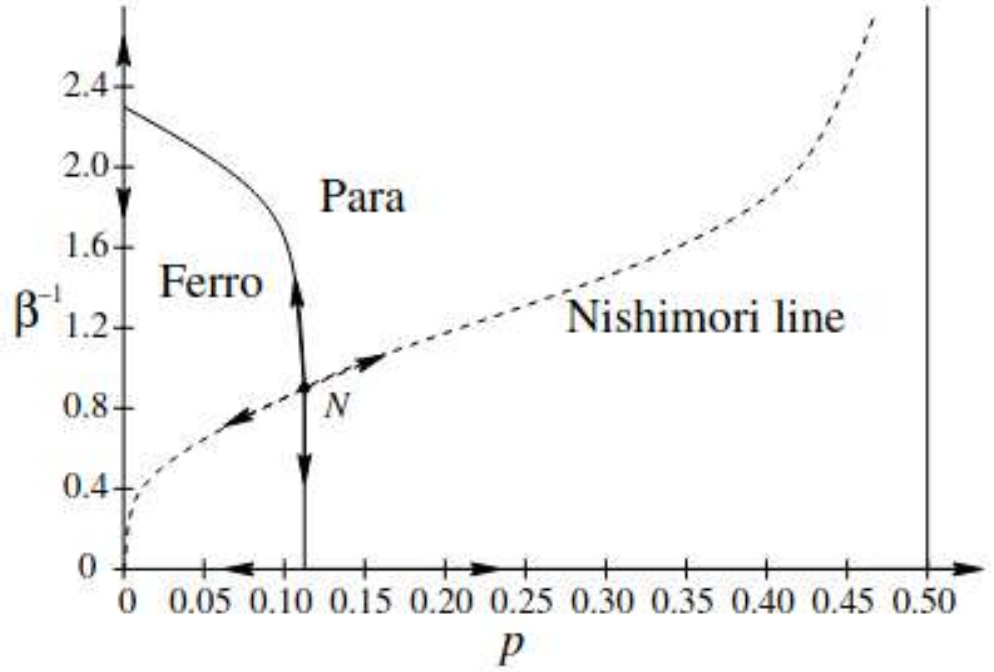
0 Temperature error correction algorithm corresponds to MWPM, is not the optimal, but close, Nishimori also proved that

the phase boundary is vertical after “N”, so

MWPM and optimal strategy (which can be

Found via monte-carlo simulation) have same

pc.



Error Correction Strategy

- Additional Ancilla qubits are used to measure syndrome
- Measure stabilizers → obtain syndrome (locations of defects)
- Defects are considered as boundaries and using MWPM algorithm we find the most probable errors strings
- Minimize the sum of free energy of each topology class
- Crucial principle:
- Only need to recover **correct topology**, not exact error
- Problem approximates to: Finding minimum-length chains consistent with syndrome

Toward Self-Correcting Quantum Memory

- In 2D:
  - Errors form strings → low energy cost → not self-correcting
- In higher dimensions (e.g., 4D):
  - Errors become higher-dimensional objects
  - Errors are surfaces , which doesnot need classical non local operation for correction
  - Energy correction can be done only using Pauli, CNOT, Toffoli gates
- Leads to:
  - Passive stability at finite temperature
- Suggests a path toward **intrinsically stable quantum systems**

References

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